

MAINS MASTER PROGRAM (MMP) 2024

ENVIRONMENT-3

MARINE AND COASTAL ECOSYSTEM

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2. MARINE ECOSYSTEM

- Introduction

- » Since oceans cover more than 75% of the earth, healthy oceans signify a healthy planet. The ocean has rich biodiversity which supplies humans with essential ecosystem services like food, employment etc. Globally the market value of coastal and marine resources and industries is estimated at US\$ 3 trillion per year nearly 5% of global GDP.
- » But, Marine life throughout the world is being subjected to immense pressure due to our perennial yearning for 'progress' and comfort and our utter carelessness in dealing with affairs of the oceans, and of land too.

- Key Factors Harming the Ocean and their impact:

- » **Climate Change:** It is causing sea level rise, ocean acidification, and changes in ocean current; This is having negative impact on marine ecosystem and marine biodiversity
- » **Marine Pollution:** It is being caused by Marine litters, industrial discharge, agricultural run off, oil spills, shipping industry, sediment flow etc. This pollution is not only harming marine life but is also polluting marine food.
- » **Overfishing:** It is leading to imbalance in marine ecosystem and disruption in the food web. This not only hampers marine biodiversity, but also hampers economic output of the economy.
- » **Ocean Degradation:** Destructive fishing practices, deep sea dredging, destruction of coral reefs etc damage the marine habitat.
- » **Invasive Species:** They are changing the biodiversity by outcompeting the native species.
- » **Deep Sea Mining:** Once this starts, it will also cause damage to marine beds and thus damage the ecosystem.
- » **Noise Pollution:** Through underwater equipment, seismic surveys etc may disrupt marine mammal communication.

- Some international initiatives for marine ecosystem:

- » **Paris Agreement** and other efforts to fight climate change
- » **High Seas Treaty, 2023**
- » **IMO Conventions:**
 - **MARPOL** (The International Convention for the Prevention of Pollution from ships) focuses on preventing pollution from the ships by operational or accidental causes.
 - **Ballast Water Management Convention, 2004** (entered into force in 2017)
- » **Lisbon Declaration** on Ocean Conservation aims to prevent, reduce and control marine pollution.
- » **Efforts to fight plastic pollution:** For e.g. Ban of several single use plastics in EU and India.
- » **FAO's Code of Conduct** for Responsible Fisheries

- Way Forward: What more could be done?

- Fighting Climate Change:**
 - More ambitious Paris Targets
 - **Lifestyle for Environment:** Promoting environment conscious lifestyle among individuals and communities
- Fighting Pollution:**

- Understanding the source, distribution, and dynamics of various pollutants and ensuring regular beach cleanup campaign.
- **Strict regulation** on dumping of waste, and other chemicals
- **Precision Agriculture** to reduce flow of excessive nutrients in soil
- **Using eco-friendly chemicals** for household cleaners
- **Reducing air pollution and water pollution on land** to reduce marine pollution and acidification of ocean .
- c. **Preventing Overfishing:** Setting up science-based catch limit; combating illegal fishing; and supporting small scale fishery sector.
- d. **Restoring Degraded Habitats:** Regrowing Mangroves; Restoring corals through initiatives like Bierocks.
- e. **Fighting introduction of invasive species:** Stricter implementation of ballast water convention; Proper regulation of aquarium trade; develop and implement targeted removal programs
- f. **Detailed EIA** should happen before beginning Deep Ocean Mining
- **Conclusion:**
 - » Ironically, much of the stress on the world's ocean originates from the land. There is a lot we too can do to prevent the burden on our oceans.

3. LISBON DECLARATION ON OCEAN CONSERVATION

- **Why in news?**
 - » **All 198 members** of the UN have unanimously adopted the Lisbon Declaration on Ocean Conservation in July 2022 on the last day of the UN Ocean Conference 2022.
- **Details**
 - » Participants agreed to work on preventing, reducing and controlling marine pollution. The nations committed to follow science based and innovative action on an urgent basis.
 - » They also agreed that developing countries (particularly small island developing states) and LDCs need **assistance with capacity building**.
 - » Developing and promoting **innovative financing solutions** to help create **sustainable ocean-based economies** as well as expanding nature-based solutions to help conserve and preserve coastal communities.
 - » Member nations also committed to empowering women and girls, recognizing their participation is crucial to building a sustainable ocean-based economy and achieving the UN-mandated SDG14.
- The conference has set the stage for the fifth session of the intergovernmental conference on an international legally binding instrument for the conservation and sustainable use of marine biological diversity of areas beyond national jurisdiction.

4. HIGH SEA TREATY (AGREEMENT UNDER UNCLOS ON CONSERVATION AND SUSTAINABLE USE OF MARINE BIODIVERSITY IN AREAS BEYOND NATIONAL JURISDICTION)

- **Introduction:**
 - » The High Seas are open ocean areas that are outside the jurisdiction of any country. It consists of around 64% of the ocean surface and around 43% of the earth. The high sea is very rich in marine species and also holds trillions of micro-organisms.
 - » There are some international legal frameworks like UNCLOS and Antarctic Treaty System which focus on protection of biodiversity, but they are not comprehensive enough.
- **On 19th June 2023**, UN Adopted the High Seas Treaty. This Treaty will work as an implementation agreement under UNCLOS, much like Paris Agreement under UNFCCC.
- **Key Highlights of the Treaty:**
 - » The treaty aims to address critical issues such as the increasing sea surface temperatures, exploitation of marine biodiversity, overfishing, coastal pollution etc.
 - » **Four Main Objectives:**
 - **Demarcation of Marine Protected Areas (MPAs):** MPAs can be established by "3/4th of majority vote". This would prevent decision to get blocked because of one or two countries. The treaty also provides for implementation, monitoring and review of MPAs.
 - **Sustainable use of Marine Genetic Resources** and **equitable sharing of benefits** arising from them: The treaty seeks to ensure that any benefits arising from genetic information of biodiversity in high seas, including monetary gains, are **free from IPR controls** and are equitably shared amongst all. Such knowledge should remain openly accessible.
 - **Obligation to conduct EIAs** for activities with potential impacts on the high seas: The treaty includes a **new impact threshold** to trigger a screening process, which means more activities will now be subject to at least some assessment.
 - **Capacity building and Technology transfer:** From advanced economies to developing and especially small island states which may not have enough resources or expertise for conservation efforts or EIA.
 - » **COP**, which acts as the decision making body of the treaty, will take the work forward and will also act as a platform to work with existing authorities that regulate fishing, shipping and mining.
- **Key contentious issue which delayed the signing of the treaty:** Many Developed countries opposed the treaty as they stand by private entities which are at the forefront of advanced R&D in marine technology.
- **Difficult road ahead:**
 - » **Treaty needs to be ratified by 60 countries** for it to come into force.
 - As of March 2024, only two countries had ratified it. (Note: India has not signed the treaty yet)
 - Note: It took 12 years for UNCLOS to get enough ratification and come into force.
 - » **The details of major contentious provisions**, including EIA, benefit sharing, mobilization of funds etc are still to be worked upon. Many other issues remain unaddressed including the mechanism for policing and protected areas.
- **The Blue Leaders High-Level Event on Biodiversity Beyond National Jurisdiction** was held in Belgium in March 2024, to urge nations to ratify the treaty. The countries have decided to use the UN Ocean Conference to be held in 2025 as a platform to bring the BBNJ Treaty into force.

5. MARINE PLASTIC POLLUTION/LITTER POLLUTION

- **Introduction: What is marine litter?**
 - » It's any man-made, long standing solid material that humans have incorrectly disposed of and that has ended up on the beach, in estuaries, rivers, seas and ocean. Plastic is the most common type of litter found at sea.
- **Reasons for Increasing Marine Litter:**
 - » Continuously growing quantity of the litter and debris disposed in oceans due to increased population, industrialization, single use plastics etc. Around 8 million tonnes of plastics end up in the world's ocean every year.
 - » Very slow rate of degradation of litter items, mainly plastic.
- **Harmful impacts**
 - » **Health:** Affects public health (plastics have now been found in human blood).
 - » **Environment:** Threatens marine ecosystem
 - Animals get trapped in this litter. They also sometimes confuse marine litter with food. It is estimated that more than 1 lakh turtles and marine mammals die every year due to plastic marine litter.
 - Ghost Fishing: Nets, Fish Aggregation Devices (FAD) and other gears continue to fish for decades after getting discarded.
 - » **Economy:** Impacts fishery and tourism sector
- **Key steps taken by India:**
 - » **Marine Plastics Survey Program of NCCR (National Centre for Coastal Research):** It found that 50% composition of marine litter was by single use plastics from 2018 - 2021 at various beaches of India
 - » **2021 Amendment to Plastic Waste Management Rules, 2016:** Ban on several single use plastics; increase in minimum thickness of plastic bags.
 - » **EPR Guidelines** related to Plastic packaging materials.
 - » **Swatch Sagar Surakshit Sagar:** Commemorating the 75th year of India's independence, a coastal cleanup drive was carried out at 75 beaches across the country across 75 days over 7500 km long coastline. This unique first ever national campaign culminated on "International Coastal Clean-Up Day" on 17th Sep 2022.
 - » At UN Ocean Conference in Lisbon, India has assured the world community that under PM Modi, it is committed to protecting at least 30% of our lands, waters and oceans, and thus adhere to its commitment of 30X30 by 2030 in a mission mode.
 - » **International Cooperation:** Under the Commonwealth Litter Program (CLiP), the UK's Centre for Environmental Fisheries and Aquaculture Sciences (CEFAS) and India's National Centre for Coastal Research (NCCR) launched a pilot project to understand deteriorating sea water quality due to marine litter.
- **Key International Initiatives:**
 - » **London Dumping Regime** (of International Maritime Organization): it regulates deliberate dumping of plastic waste at sea from vessels and platforms.
 - » **International Convention for the Prevention of Pollution from Ships (MARPOL)**
 - » **High Seas Treaty, 2023**

- **Way Forward:**
 - » Understanding the source, distribution, and dynamics of plastics is important to target priority areas.
 - » Regular beach cleanup campaigns.
 - » Other steps discussed with marine conservation
 - » Enforce international Conventions.
 - » More R&D - focus on plastic degrading enzymes - this can provide promising future prospect in managing and recycling robust plastics such as Polyethylene Terephthalate (PET).

6. OCEAN DEOXYGENATION -> DEAD ZONES

- **Ocean Deoxygenation**
 - » It is the phenomenon of oxygen loss in ocean caused by excessive growth of algae due to nutrient pollution. The nutrient pollution may be caused by fertilizers, sewage, animal or aquaculture waste.
- **The IUCN Report 'Ocean deoxygenation: Everyone's problem': Key Findings**
 - » **Ocean regions with low oxygen concentration** have expanded to all depths of the Ocean
 - » The **volume of area depleted with oxygen**, known as "**anoxic waters / dead zones/ biological deserts**" have quadrupled.
 - » **What if the present situation continues?**
 - Business as usual approach -> Ocean to lose 3-4% of oxygen by 2100; Local changes will be more severe.
 - Further, most of the oxygen loss will take place in the upper 1,000 meters, which is the richest part of the Ocean for biodiversity.
- **Reasons for Ocean Deoxygenation:** In recent years, anthropogenic factors have become the main cause:
 - » **Nutrient Pollution -> Eutrophication -> Oxygen depletion**
 - » **Climate Change** -> Warm water has lower oxygen dissolved; further warm water increases metabolism rate and thus the oxygen demand.
- **Adverse impact of low oxygen levels**
 - » **Balance of Marine Life:** The IUCN report has started impacting the balance of marine life, favoring species tolerant of low-oxygen conditions, like jellyfish, some squid and microbes, at the expense of species sensitive to low-oxygen, including most fish and many marine species.
 - » Negatively hamper cycles of nitrogen and phosphorus
 - » **Negative Economic Impact:** Decline in fishery can harm coastal communities
 - » **Poor Aesthetics:** Ocean covered with algae and weed negatively hamper tourism.
 - » **Intensification of Climate Change:** No photosynthesis in dead zone, release of N₂O etc can intensify climate change.
- **Way Forward**
 - » **Tackle Climate Change:** There is an urgent need **to raise our ambitions** to tackle and immediately curb our emissions.
 - » **Nutrient Management:** Ensuring optimal utilization of fertilizers in agriculture to reduce the amount washing off to oceans - Precision Agriculture
 - » **Fighting Marine Pollution** - already discussed

7. PROTECTION OF COASTAL REGION

- Introduction

- » Coastal zones are places of enormous ecological, cultural, social and economic significance. They contain unique and sensitive ecosystem of great natural and economic value and is home to numerous endangered species. The region also serves as home to 50% of the world's population and generated 40% of the global economic activities.

- Key Problems Faced by Coastal Regions:

- » **Climate Change:** Along much of the earth's coast a **warming climate and sea level rise** are already negatively affecting natural ecosystems and human communities
- » **Rapid Industrialization and Deforestation**
- » **Pollution** due to mining, municipal waste disposal and industrial waste disposal are also leading to environmental problems in coastal regions.
- » **Invasive Species** -> Biodiversity loss
- » **Coastal Erosion** has started hampering a number of coastal regions. E.g. Vishakhapatnam

1) COASTAL REGULATION ZONE

- CRZ notification is issued under the **Environmental Protection Act, 1986** for regulation of activities in the coastal area by the MoEF&CC.
- It classifies the coastal land upto 500 m from the HTL as the Coastal Regulation Zone (CRZ). The **CRZ** are further classified in **four categories**:
 - » **CRZ-1** are ecologically sensitive areas.
 - **CRZ 1-A** constitute the ecologically sensitive area and the geomorphological features which play a role in maintaining the integrity of the coast viz: Mangroves, corals, sand dunes, salt marshes, national parks, WLS, Reserved forests, nesting grounds for turtles, birds etc.
 - **CRZ 1-B** consist of inter-tidal zones (between HTL and LTL)
 - » **CRZ-2** are areas that have been developed upto or close to the shoreline. Unauthorized structures are not allowed in this zone.
 - » **CRZ-3** are areas that are relatively undisturbed (both urban and rural)
 - » **CRZ-4** are areas covered between Low Tide Line and 12 nautical miles seaward
- **CRZ Notification, 2018: Easing of Norms for CRZ:**
 - » **Need:** Stringent norms were hindering development activities;
- **Key Changes**
 - » **Delegation of Project Clearance Power to State Governments.**
 - **Only Projects in CRZ-1 and CRZ-IV will require permission from Union Ministry.** The Powers to clear projects in **CRZ-2 and CRZ-3 have been delegated to State Governments**
 - » **Defreezing of Floor Area Ratio (FAR) in construction norms**
 - Earlier, for CRZ-2, it was frozen to 1991 Development Control Regulation (DCR) levels, Now, it will be based on laws which are in vogue.
 - » **Relaxation of No Development Zone (NDZ) criteria**
 - Densely populated (density > 2,161 per sq km) rural areas (under CRZ-III) referred as CRZ-III-A, now have a NDZ of 50 m from the High Tide Line (HTL) as against earlier 200 meters.

- Further, for island close to the mainland coast and for all backwater islands the new NDZ is 20 m.
- » To **fight pollution**, treatment facilities have been made permissible activity in CRZ-I B area, subject to necessary safeguards.
- » **Steps to Facilitate Tourism:**
 - Permission of temporary tourism facilities such as shacks, toilet blocks, change rooms, drinking water facilities etc, in beaches even in the NDZ of the CRZ-III.
- **Analysis: Positives**
 - » **Economic growth** through promotion of tourism by ensuring more activities, more infrastructure and more opportunities in the coastal region.
 - » The **reduction in NDZ** will also help government achieve infra objectives such as that of PMAY.
 - » In sync with the **port-led industrialization** approach.
- **Analysis: Criticism**
 - » **Climate change adaptation** should require us to move away from coastal region, not go near it.
 - » **Hampers fishermen livelihood:** Promotion for infrastructure for tourism around coastal regions, not only make tourists vulnerable to disasters, but also hinders fisherman from earning their daily livelihood.
 - » **Significant cost to environment:** Infrastructure development may not only harm the biodiversity of the region, but destruction of sand dunes, forests may also make Indian coasts more vulnerable to Tsunami and other natural disasters.

2) COASTAL AQUACULTURE AUTHORITY (AMENDMENT) ACT, 2023

- Why in news?
 - » Coastal Aquaculture Authority (Amendment) Act, 2023 passed by both houses of the Parliament
- **Background:**
 - » The Coastal Aquaculture Authority Act 2005 was enacted with an aim to protect coastal environment, while promoting orderly growth of coastal aquaculture farming in coastal areas.
 - » It has established Coastal Aquaculture Authority which regulates activities connected with coastal aquaculture in the coastal areas.
 - » It also defined coastal aquaculture to include culturing under controlled conditions in ponds, pens, enclosures or otherwise, in coastal areas of shrimp, prawns, fish or any other aquatic life in saline or brackish water, but doesn't include freshwater aquaculture.
 - » It ensures continued operation of coastal aquaculture within **CRZ area** subject to restrictions imposed by the Authority.
 - » It also penalizes unregistered farms in prohibited areas.
- **Impact:**
 - » **Facilitated millions of jobs, self-employment opportunities, businesss** and environment protection.
 - » **Increased production of fishery sector.**
- **Need of Amendment:**
 - » Some ambiguities related to the provisions of CRZ notifications - like that of "**No Development Zone**" has been misinterpreted to be applicable in the hatcheries as well.
- **2023 Amendment:**

- » **Broadens the definition of Coastal aquaculture** to include things like cage culture, sea-weed culture, bivalve culture, marine ornamental fish culture etc.
- » **Registration given under the Coastal Aquaculture Authority Act** will be considered a valid permission under CRZ notification. This will enable lakhs of small marginal aquaculture farmers to avoid the possible need for obtaining CRZ clearances from multiple agencies.
- » **Some Aquaculture activities** like hatcheries, nucleus breeding centres etc can be established in NDZ [200 m from HTL] of seas and buffer zones of creeks/rivers/backwaters.
- » **Decriminalization** of cases of illegal coastal aquaculture and fixed penalties.
- » **Increase in role of the Authority:**
 - Fix standards of input and discharge of effluents from aquaculture units.
 - Prohibition of certain harmful inputs
 - Monitor and regulate various aquaculture units, inputs and outputs.
- » The act also **prohibits the use of insecticides** and other pharmacologically active substances that can harm human health in coastal aquaculture.

3) COASTAL EROSION

- PYQ:
 - » Explain the causes and effects of coastal erosion in India. What are the available coastal management techniques for combating the hazard? [Mains 2022, 15 marks, 250 words]
- **Introduction:**
 - » A coastline is a complex series of interlinked physical systems in which both offshore and onshore processes are involved. Coastal Erosion is one of these physical processes. It refers to wearing away of land and the removal of beach and dune sediments by wave action, tidal currents, drainage or high winds.
- **Causes of Coastal Erosion** can be divided into two broad categories: **Natural** or **Manmade**:
 1. **Natural Causes:**
 - i. **Wave Action** is the main cause of coastal erosion.
 - ii. Winds, tides, near shore currents, sea level rise etc. are also causing erosion
 - iii. **Phenomenon of subsidence** is a regional phenomenon that lowers the surface area in a specific region.
 - iv. Catastrophic events like severe storms, tidal surges, and cyclones can cause severe erosion.
 2. **Manmade Factors:**
 - i. **Infrastructure creation in coastal regions:** For e.g., building houses via land reclamation or within sand dune areas.
 - ii. **Sand removal** above replenishable quantities from the coast upsets the longshore sand transport budget and can result in erosion
 - iii. **Coral Mining and other means of damaging protective corals** may cause beach degradation.
 - iv. **Structures like seawalls, breakwaters** also have a side effect as it increases erosion of adjacent areas.
 - v. **Deforestation:** Damaging of mangroves and other coastal vegetation is a major factor.
 - vi. **Climate Change** which is mostly human induced is leading to sea level rise which is eroding more and more coastal regions.
 - vii. **Unscientific Coastal Management**
- **Impact of Coastal Erosion**
 - Floods including worsening impact of high tide flooding.
 - Saltwater penetration into rivers, coastal agriculture plains
- **Coastal Erosion Control Strategies:** There are **three coastal erosion control methods.**
 - A) **Soft Erosion Controls/ Non-Structural Methods:** These methods are **temporary options of slowing the effects of erosion.**
 - **Artificial nourishment** of beaches
 - **Coastal Vegetation** such as mangrove and palm plantation

- **Dune Reconstruction/rehabilitation**
- Other options are **beach scraping** and **beach bulldozing** which allows for the creation of artificial dunes in front of building or as means of preserving building foundation.
- Most common method is the **Beach nourishment** projects.
 - It involves placing **additional sand on a beach** to serve as a buffer against erosion or to enhance the recreational value of the beach.
 - Because nourishment doesn't stop erosion, it has to be repeated to maintain the beach.
 - **Advantages**
 - Restores and widens recreational beach
 - Structure behind the beach is protected as long as the added sand remains
 - Beach nourishment doesn't leave hazards on the beach or on the surf zone.
 - **Disadvantages**
 - Erodes faster than natural sand so **continuous refurbishing** required.
 - Number of Storms affecting the beach makes the life time of the nourishment vary.
 - Expensive, and must be repeated periodically
 - Process of nourishment may damage, destroy or otherwise hurt marines and beach life by burying it, squishing it under bulldozers, changing the shape of the beach, or making the water near the beach too muddy.
 - Difference in "grain size" of the added sand affect the way waves interact with beach. This will affect surf conditions and bars on the submerged part of the beach.

B) **Hard Erosion Controls/ Structural Measures**

- More permanent solution than soft erosion methods.
- **Seawalls and groynes(or groin)/breakwaters** serve as permanent infrastructure; Tetrapod-based seawall are also included in the category.
- **Limitations**
 - Not immune from normal wear and tear and will need **refurbishment or rebuilt**.
 - Further, as the understanding of natural shoreline function improves, there is a growing acceptance that structural solution can cause more problems than they solve. It interfere with natural water currents, and prevent sand from shifting along coastline.
 - They also cause erosion to adjacent beaches and dunes and lead to unintended diversion of stormwater and waves onto other properties.

C) **Combination of the Structural and Non-Structural Methods** (i.e. combination of hard erosion control and soft erosion control)

- This hybrid method reduces limitations of both the methods and provides better efficacy and efficiency.
- Some of the common approaches of combination are:
 - a. **Combining Beach nourishment** with artificial headlands/groynes.
 - b. **Revegetation with temporary offshore breakwaters/ Artificial reefs** is commonly used.

- Using a combination of beach nourishment and groynes/artificial headlands promotes the trapping of the downdrift movement of the sediments, thus reducing downdrift erosion. This also reduces the frequency of re-nourishment.
- D) **Relocation**
 - Human beings move away from the coast and surrender the coast to natural processes of both absolute and relative sea level rise and erosion. This solution is eco-centric meaning that the focus is on forcing humans to adapt to natural processes rather than the opposite.
- **Situation of Coastal Erosion in India (April 2022)**
 - **Ministry of Earth Sciences** has informed the Lok Sabha that about 34% of coastal region in India is under varying degree of erosion.
 - Of the rest, 40% is stable and 26% is accreting in nature.
- **Other Way Forward:**
 - **Upgrade infrastructure** - raising of roadways, installing backflow preventors, increasing distance between the first floor and the base flood levels.
 - **Create Resilient Shoreline: Preserve and restore coastal habitats** - Communities should work with nature to conserve and restore marshes and barrier islands.

4) COASTAL REGION AND CLIMATE CHANGE

- Climate change has emerged as the biggest threat to the future of human civilization. The coastal communities also face enormous challenges from natural calamities and livelihood loss caused by climate change.
- **Keyways in which Climate Change impact Coastal communities and Coastal Areas:**
 - **Sea Level Rise:**
 - Globally sea level is rising at the rate of 4.5 mm per year. This sea level rise is causing historical shore line changes and wave climate changes that engulf landmass with economically potential physical infrastructures. Coastal communities and other stakeholders are also affected by loss of land, erosion, flooding, and saltwater intrusion in coastal aquifers. This causes ecosystem degradation, reduction in potable water, an increase in hazards and health ailments, a rise in homelessness and poverty, involuntary migration, loss of employment, and an increase in debt.
 - **Increased Sea Surface Temperature (SST):** Increasing SST affects fish migration, fish physiology, fish breeding, fish recruitment, and habitat loss. It also enhances ocean acidification, salinity, and longshore current patterns which impact primary production and fish stock in the area. Various studies have shown that it has impacted distribution and diversity of fish groups in coastal area of India.
 - SST also intensifies cyclones and increase the extent of damage caused by it in coastal region.
 - **Shoreline Change (SLC):** Study undertaken by NCSCM utilizing 28 years of shoreline data show that 33.6% of India's cost is eroding. It impacts livelihood, property, recreation, tourism, etc.
 - **Frequency of Cyclones and Floods:** During the 21st century, there has been an increase in the occurrence and severity of flood hazards in India. Cyclones and floods cause casualties, and

injuries besides the devastation of coastal infrastructure, roads, schools, hospitals, houses etc. In addition, health hazards due to injuries and epidemics enumerate as common problem due to floods, storms, and cyclones.

▪ **Saltwater Intrusion:**

- During dry months, salt water intrusion takes place in wells and bore wells located near the coast.
- In addition, over-harvesting of water from aquifers, SLR etc. is also aggravating saltwater intrusion.
- This impacts productivity of agriculture and livestock; hampers availability of drinking water; increases vulnerability of coastal communities.

▪ **Drought:**

- Climate change leads to variability in rainfall and frequently cause droughts, including in coastal areas. The drought causes economic and health challenges for the local community.

▪ **Reduction in capture of fisheries:**

- Climate change leads to SST, changing current patterns and negatively impacts biology (specially reproductive biology) of the fish. It also alters habitats and migratory routes of the fish.
- Reduction in fishery hampers food security (protein, vitamin, and micronutrients) for the coastal community. It also hampers per capital income and increases conflict among fishing communities. This causes increased poverty, distress migration and thus increasing inequalities.

- **Way Forward:**

- **Identify location specific problems and vulnerable groups:** Hazard profile maps
- **Evacuation and Shelter Location** needs to be demarcated at suitable locations for people to move to at the time of flood and cyclones. This should be informed to coastal communities through awareness creation activities, capacity building activities, and mock drills.
- **Building Codes:** Building codes need to be modified and implemented to ensure flood proof and cyclone proof infrastructure.
 - Older buildings need to be retrofitted to ensure safety during floods/Tsunamis/Cyclones.
- **In drought prone regions** - Special focus on rain water harvesting; creation of water tanks; watershed development etc. has to be done.
- **Steps to maintain fish stocks:**
 - Fish stock trends and assessments shall be conducted to develop policies and schemes to replenish the economically important fishery resources with the involvement of local stakeholders
 - To reduce overexploitation, near shore cage culture, aquaculture, and mariculture activities should be promoted with the participation of local communities under PPP model.
- **Increased support for R&D on coastal climate** should be prioritized.

- **Conclusion:**

- Climate change is going to bring along with it unique challenges for the coastal region. It is important that appropriate steps are taken to mitigate and adapt to these challenges.

8. BLUE ECONOMY

- Blue Economy refers to the "Sustainable use of ocean resources for economic growth, improved livelihood, and inclusive development while preserving the health of Ocean Ecosystem". In other words, it means 'sustainable ocean based economy' or 'greening of Ocean Economy'.
- BE straddles the principles of marine led growth; protection of marine ecosystem; and enhanced maritime security in all national and regional manifestation.
- **Blue Economy Situation in India:**
 - » India's Blue Economy concept is multi-faceted and plays an important role in the country's economic growth because of its enormous maritime interest. It accounts for roughly 4% of the GDP and is estimated to increase once various hurdles are removed.
 - » **India's Blue Economy Sectors** include:
 - Fishing.
 - Marine Biotechnology.
 - **Mining:** The two mineral deposits of commercial significance to developers in the Indian Ocean are Polymetallic nodules and Polymetallic Huge sulphides.
 - **Offshore Wind Energy; tidal energy etc.**
 - **Shipping and Shipbuilding:**
 - These are also important aspects of the blue economy in India. The modal share of coastal shipping has the potential to increase to 33% by 2035, up from roughly 6% presently.
 - Most of the countries oil and gas is supplied by sea, leading to the Indian Ocean Region being critical to India's economic growth.
 - **Global Economic Corridor:** The Indian Ocean has become a global economic corridor. India also has significant diplomatic interests in the Indo-Pacific. Further, it has international commitments in the region under the UNCLOS, such as Search and Rescue, Seabed mining, and counter-piracy.
 - Marine tourism and leisure.
 - Education and R&D etc.
- **Key Government Initiatives to leverage Blue Economy for the benefit of the country:**
 - The Ministry of Earth Science has released the draft of National Policy on Blue Economy in 2021. The goal is to increase the blue economy's contribution to India's GDP, improve the lives of coastal residents, protect marine biodiversity, and ensure national security of maritime areas and resources.
 - There have been **efforts to boost fishery sector** in a sustainable way:
 - Integrated Development and Management of Fisheries Centrally Sponsored Scheme (CSS) was established in 2015-16 with a five-year budget of Rs. 3,000 crores (US\$ 384.3 million).
 - The 'Fisheries and Aquaculture Infrastructure Development Fund' (FIDF) was established in 2018-19 with a fund size of Rs. 7,522.48 crores (963.5 million) to provide concessional credit to state/UT governments, their entities, and the private sector to fill significant gaps in the fisheries infrastructure.

- The Pradhan Mantri Matsya Sampada Yojana was launched in May 2020 with the allocation of Rs. 20,050 crore (US\$ 2.5 billion) to bring about a Blue Revolution through sustainable and responsible development of the country's fisheries sector
- CRZ notification 2019 are also efforts to protect coastal ecosystem while promoting economic development.
- **Benefits of Blue Economy:**
 - » Poverty reduction, food security and enhanced economic opportunities
 - » Better governance of marine ecosystems
 - » Environmental benefits: Lower pollution, Biodiversity Protection
 - » Mitigating Climate Change
 - » **Synergy with SDGs:** The idea of Blue Economy is in sync with SDG Goal 14: "Life below water"
 - » Geopolitical benefits
 - » **High Return on Investment:** Recent studies have shown that for every rupee invested in Blue Economy, output can be upto five times i.e. of Rs 5.
- **Key challenges to Blue Economy for India** (Sustainable Ocean Based Economy)
 - » Majority of littoral countries are **developing countries** -> their lack of capacity, skills and access to cost-effective technologies make it difficult to achieve sustainable development.
 - » These countries host 1/3rd of the world's population and rely heavily on marine resources for livelihood and food security. This sheer pressure subject Indian Ocean resources to pressure from **pollution, habitat degradation, and over exploitation.**
 - » **Climate Change, Sea Level Rise and Ocean Acidification** are also leading to changes in distribution of aquatic species and decreased economic productivity.
 - » **Governance challenges in fishery management in Indian Ocean:** Regulations and policies that govern fisheries management and their jurisdiction are largely ineffective in mitigating **illegal, unreported and unregulated (IUU) fishing.**
 - » **Absence of effective cooperation** among South Asian countries (for e.g. SAARC's summit hasn't been held since 2014)
- **Way Forward:**
 - » **International Cooperation:** Growing competition will lead to unsustainable exploitation, whereas cooperation allows resources extraction in a more sustainable way. Organizations like **Indian Ocean Regional Association (IORA)** and **BIMSTEC** can contribute heavily in achieving the goal of Blue Economy in India's immediate neighborhood. They would required to create an improved governance framework which should create integration of scientific management and enforcement in the Indian Ocean region.
 - » **Adoption of Marine Specific Policies** in fishery, tourism, transportation etc.
 - » **Creation of infrastructure and Human Resource creation** for the future requirement of the **emerging Blue Economy:**
 - Effective implementation of **Sagarmala** initiative would improve infrastructure
 - To create more human resource, we need to introduce more courses related to ocean engineering, shipping, etc. in our Higher Educational Institutions.
 - » **More R&D in Blue Economy:** To utilize the potentially of Blue Economy fully, it is crucial to promote more R&D and take science and data based decisions.

- » **Focus on sustainability:** India's vision for Blue Economy would only succeed if environmental challenges like pollution, climate change etc. are effectively curtailed. Achieving the Paris Climate targets would be crucial here.
 - Along with mitigation, adaptation to climate change would be crucial.
- » **Strengthen Indian Coast Guard (ICG)** with more resources
 - ICG is one of the major maritime law enforcement agencies in the Indian Ocean Region. It is playing a bigger role in averting major pollution incidents, anti-poaching, and search & rescue operations. It also has significant role in anti-piracy activities.
 - The future will increase the need of these services and ICG should be enabled to deal with these issues comprehensively.
- **Conclusion:**
 - » The **Blue Economy** in India is poised for significant growth in the next few years. It's Blue Economy Mission undertaken by the government can lead to the sector becoming the next economic multiplier, depending on the execution of the policies that have been decided

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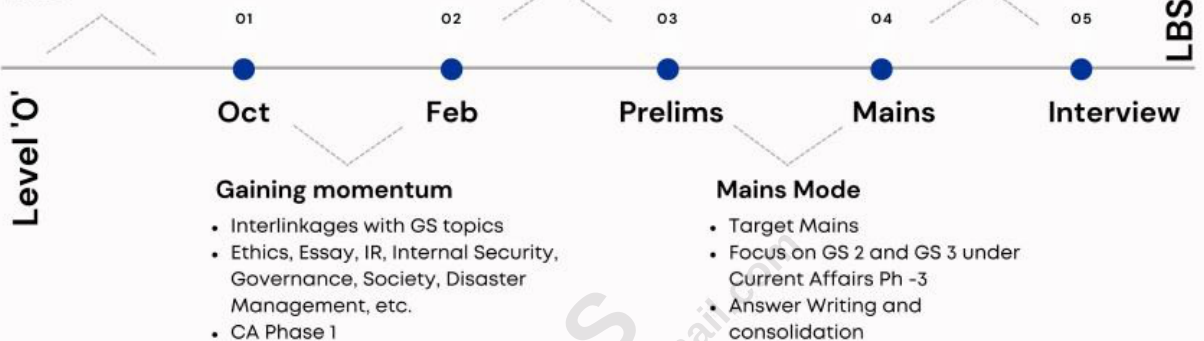
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